

Transient localization from fractionalization: weak heat conductivity in gapless spin liquids

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Introduction

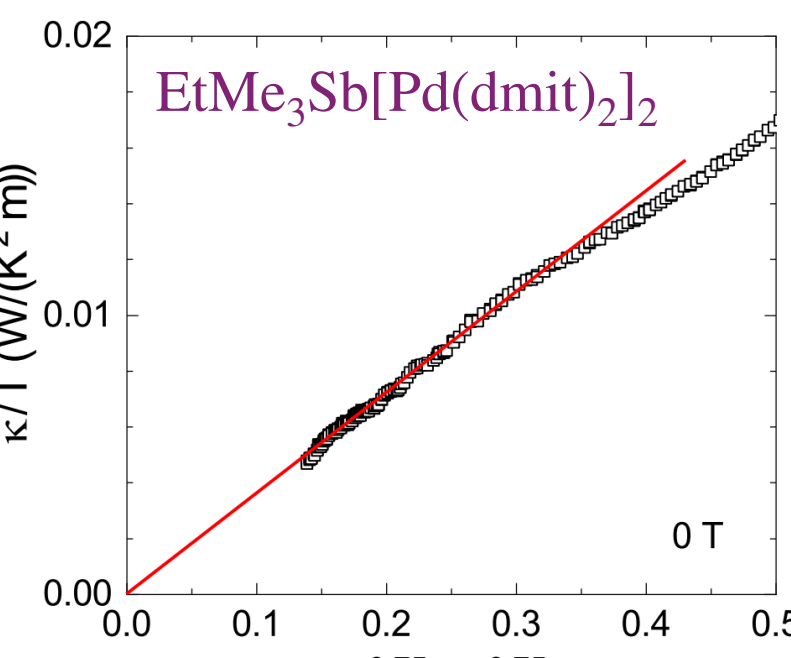
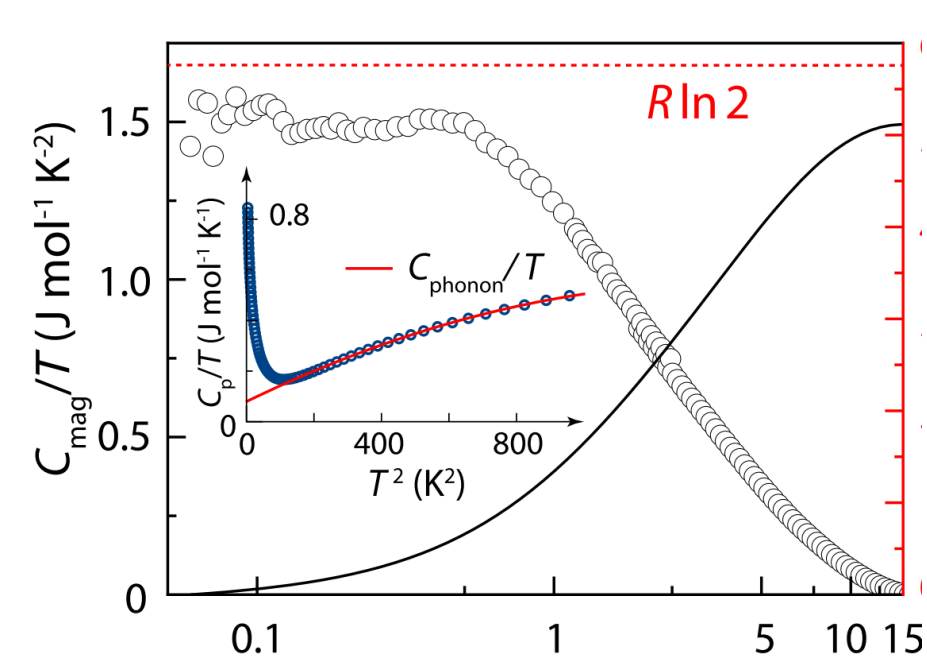
Paradox in gapless frustrated magnets

What people expect for a gapless Fermi liquid

- Specific heat of gapless fermions: $C_V \sim \gamma T$
- Residual thermal conductivity: $\kappa_0/T \sim \text{constant}$

Experimental results for gapless frustrated Mott insulators

	SmB ₆	YbB ₁₂	dmit-131	NaYbS ₂ , NaYbSe ₂	1T-TaS ₂	UTe ₂
γ (mJK ⁻² mol ⁻¹)	~ 25	~ 3.8	~ 20	~ 1.1	~ 1.84	~ 60
κ_0/T (mWK ⁻² cm ⁻¹)	~ 0	~ 0.01	~ 0	~ 0	~ 0	~ 0



Dai et al.,
Phys. Rev. X 11, 021044 (2021)
Li et al.,
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Bourgeois-Hope et al.,
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Ni et al.,
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Matsuda, Shibauchi, Kee,
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- Specific heat of gapless spin liquids: $C_V \sim \gamma T$
- While residual thermal conductivity: $\kappa_0/T \rightarrow 0$???

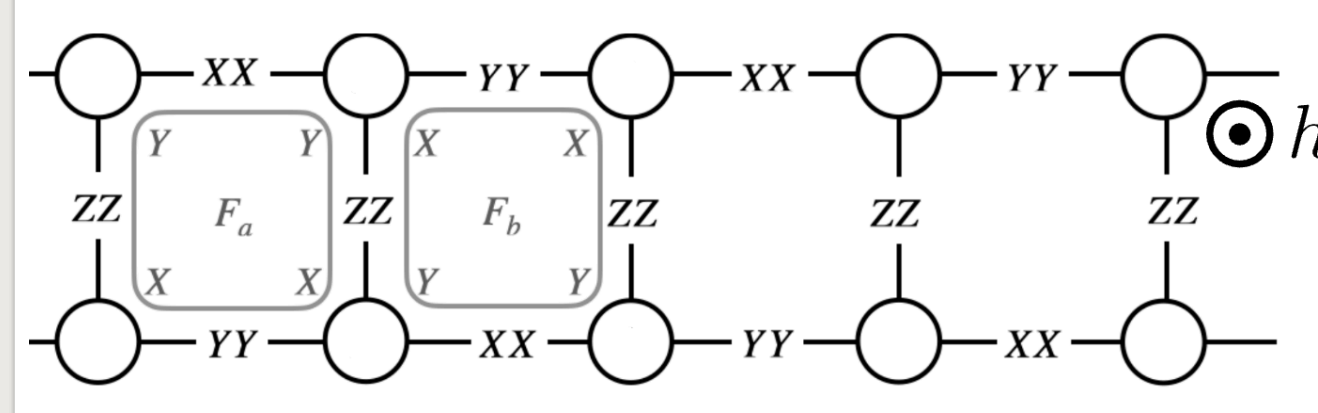
Key results

- Quantum spin liquid can self-localize without quenched or extrinsic disorder, strongly suppressing heat conductivity
- One species of fractionalized particles (visons) localizes its complementary fractionalized partner (Majorana fermions) up to an observable transient timescale
- Quasi-static visons simultaneously closes fermion gap, producing gapless spin liquids from gapped spin liquids
- Suggesting coexistence of linear C_V and vanishing κ_0/T

Disorder-free self-localization

Translation-invariant coherent disorder

A toy model for demonstration: external field induces strong vison fluctuation



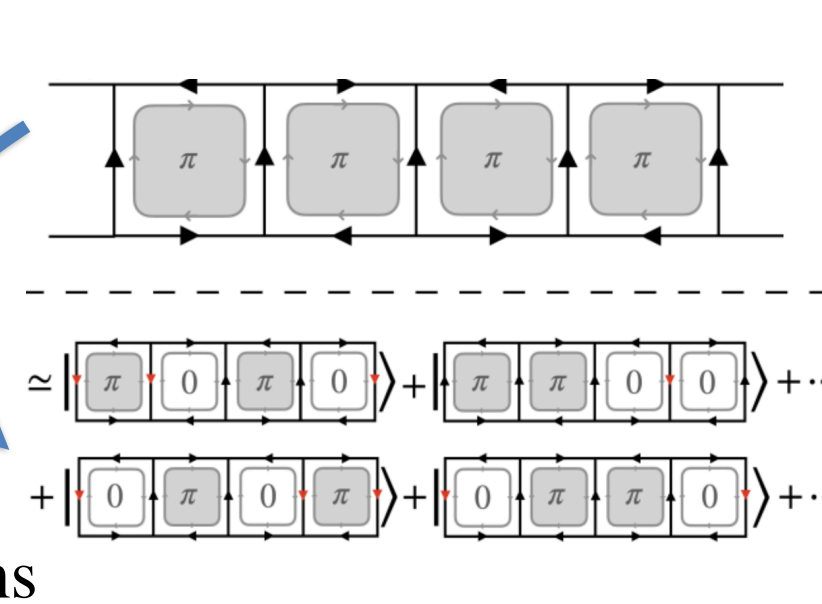
$$H = H_{\text{ladder}} - h \sum_{j,l} (X_j^l + Y_j^l + Z_j^l)$$

$$H_{\text{ladder}} = \sum_{j+l \in \text{even}} (X_j^l X_{j+1}^l + Y_{j-1}^l Y_j^l + Z_j^l Z_{j+1}^l).$$

Ground state is a π -flux ladder with free Majorana fermions

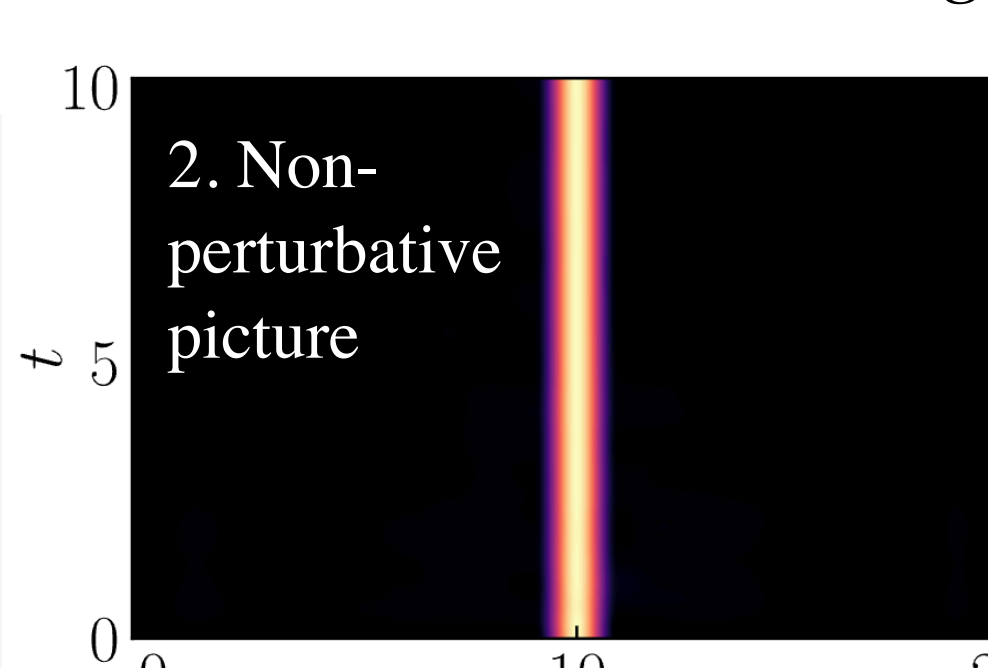
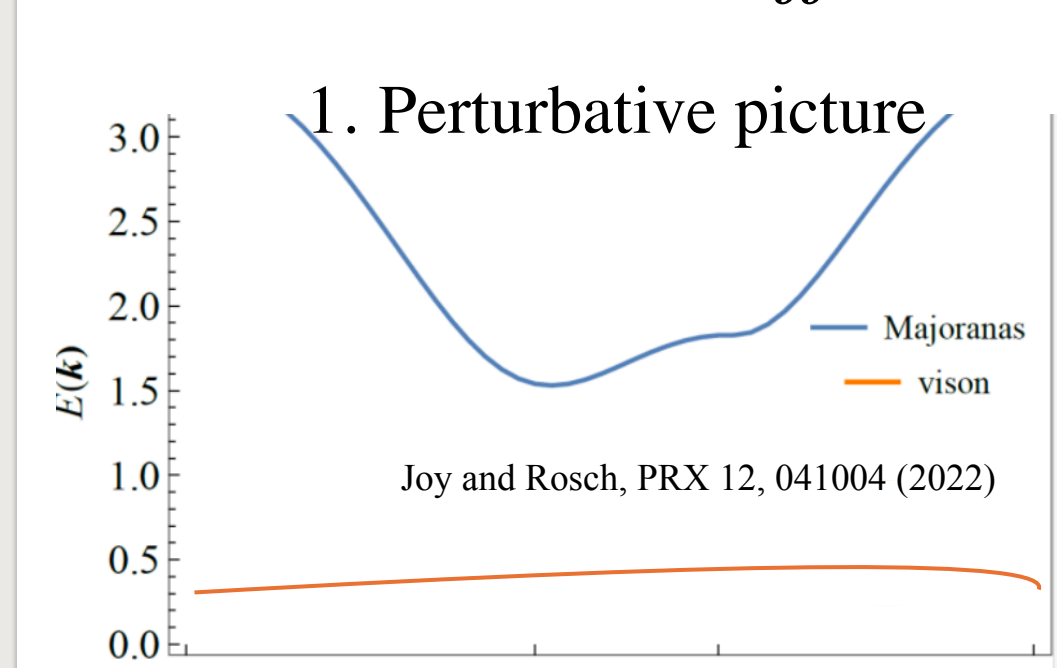
Magnetic field induces finite density of visons in the ground state

The ground state can be approximated by a superposition of vison configurations



Quasi-static visons

Field-induced effective visons mass is much larger than Majorana fermions



1. Destructive interference in vison hopping $\rightarrow$ almost flat vison band under weak field

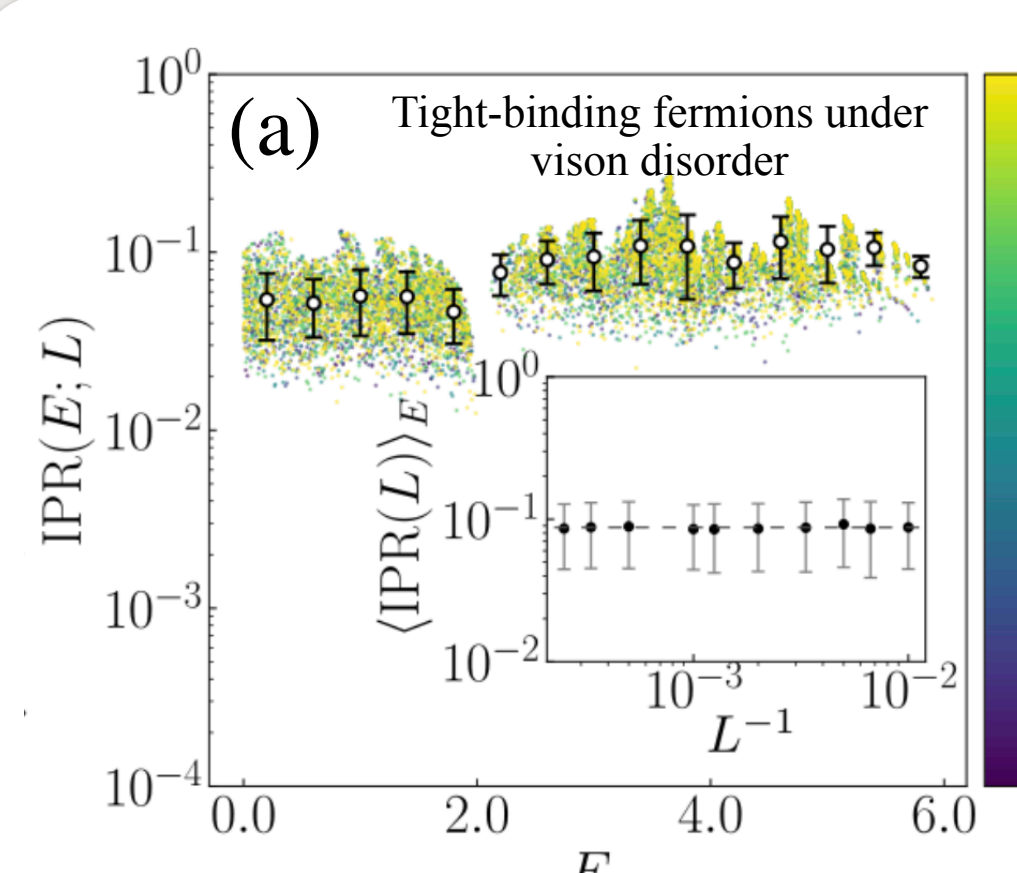
2. Quench vison dynamics (connected correlation function of vison operators)

$$C_{F,F_0}(r, t) = \langle 0 | F_r(t) F_0(t) | 0 \rangle_c$$

Evaluated under moderate magnetic field

Spinon localization without extrinsic disorder

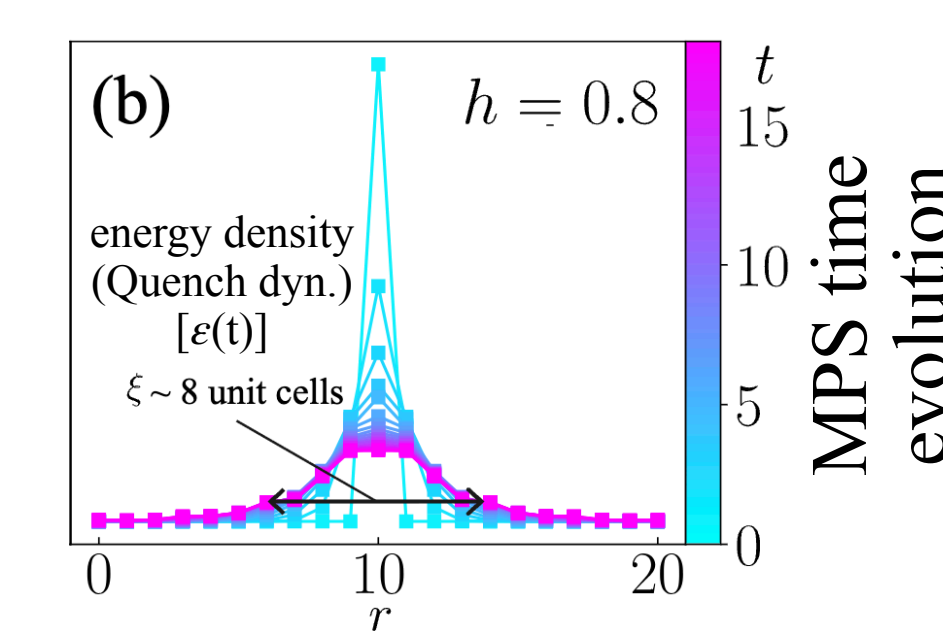
Treating visons as quasi-static, fermions are localized in the entire bandwidth



$$\text{IPR}(E, L) = \frac{1}{N(E)} \sum_{n: E_n \approx E} \sum_i |\psi_n(i)|^4,$$

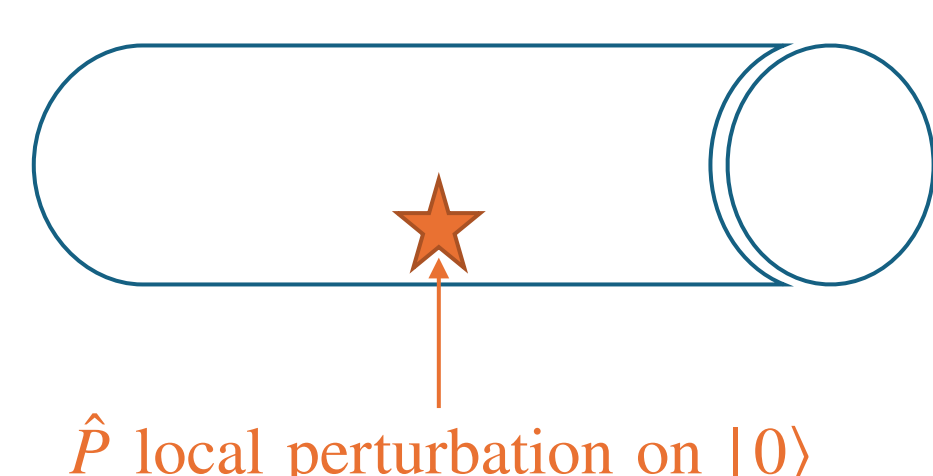
$$\langle \text{IPR}(L) \rangle_E = \frac{1}{N_{\text{tot}}} \sum_{n=1}^{N_{\text{tot}}} \text{IPR}(n, L)$$

$$\text{IPR} \sim \xi^{-1} \sim 0.1$$



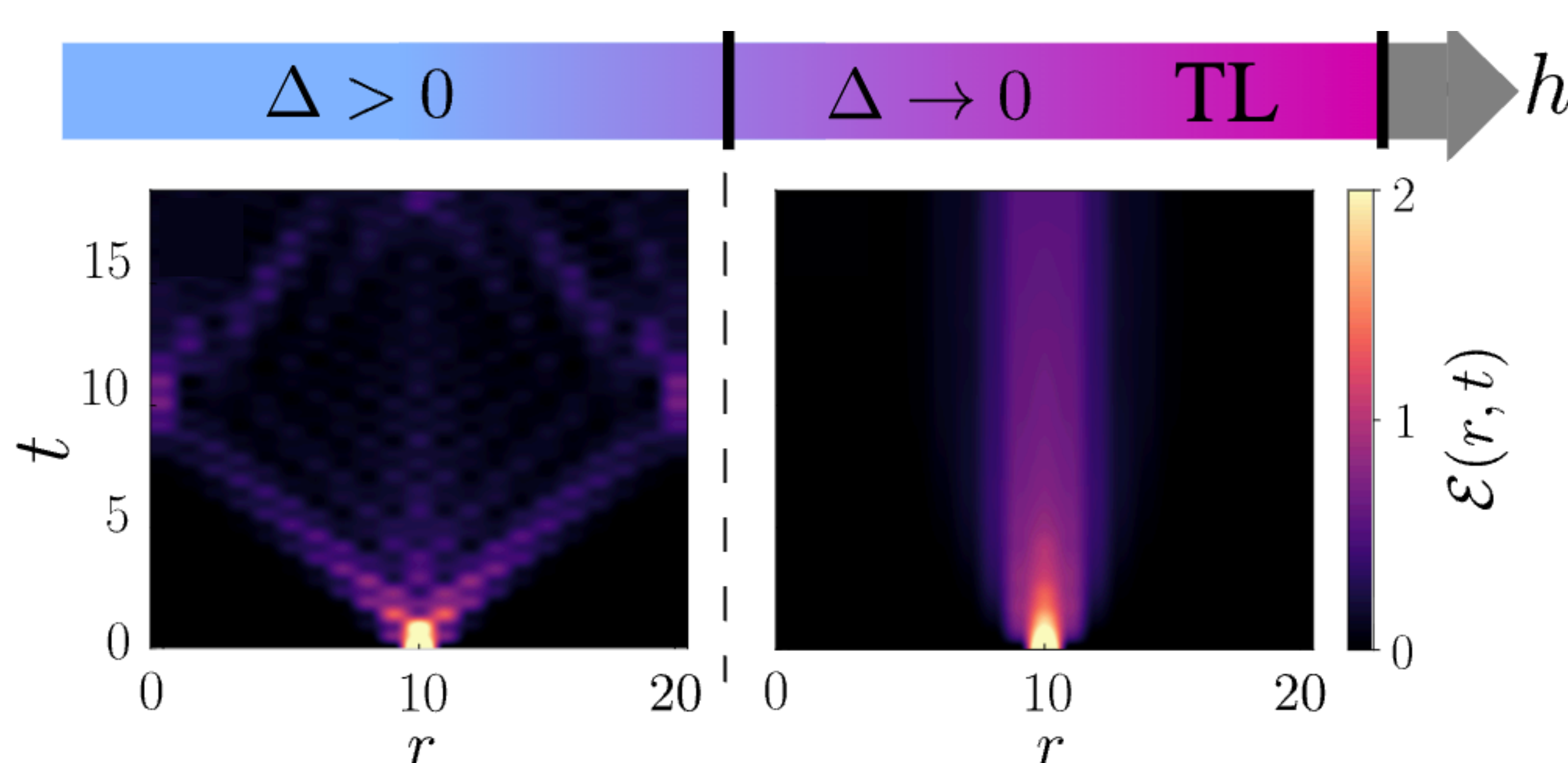
Numerical Evidence: suppressed heat conductivity and emergent gapless spectral functions

Field-induced slow quench dynamics



$$\langle 0 | \hat{P} \varepsilon(t) \hat{P} | 0 \rangle$$

Measure energy density ε spreading (MPS time evolution)



Free Majorana fermions Fast (ballistic) spreading

Transiently localized fermions Extremely slow spreading

Heat conductivity ($T \rightarrow 0$)

Energy conductivity is strongly suppressed at finitely low ω , albeit with gapless energy density response

Define energy (ε) and current operator (J) using discrete continuity equation:

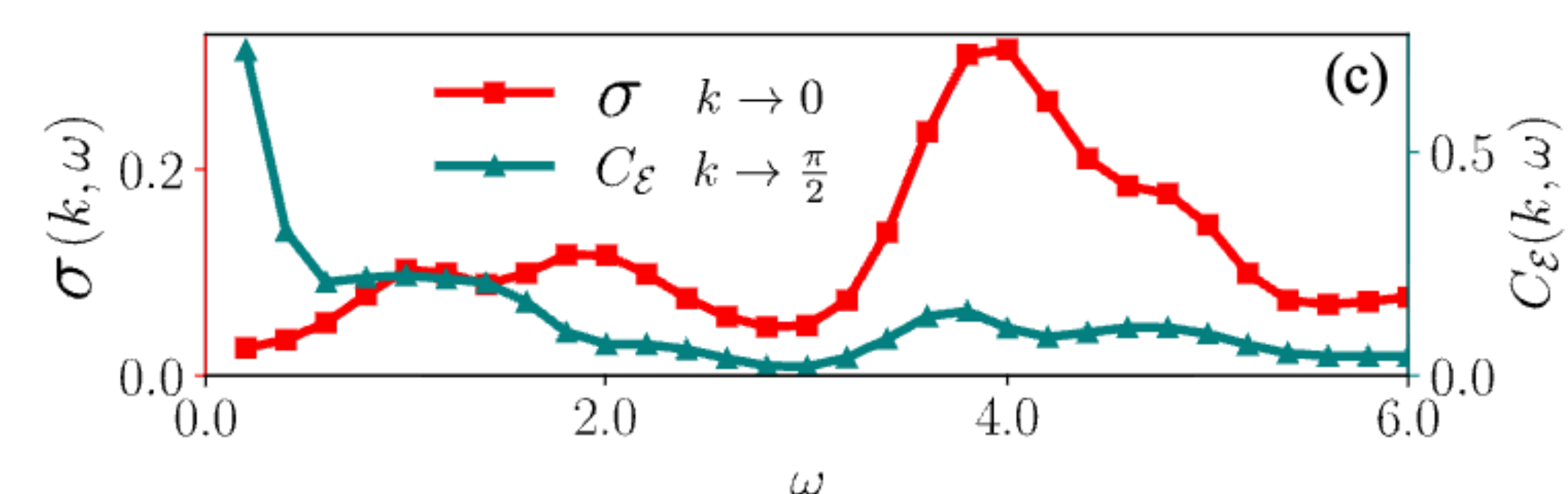
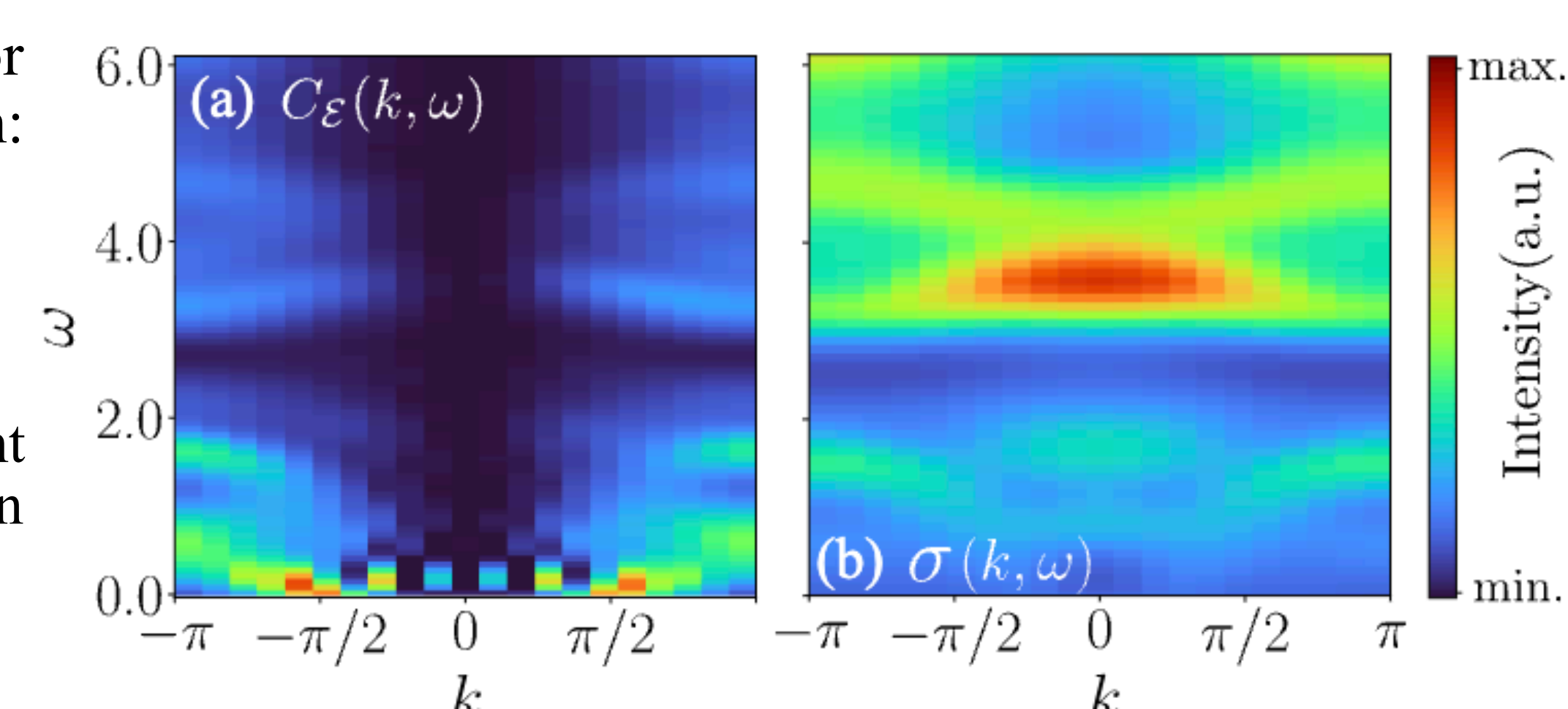
$$\frac{d}{dt} \varepsilon(r, t) + [J_{r,r+1}(t) - J_{r-1,r}(t)] = 0$$

Compute energy conductivity by current correlation or energy-density correlation Using Kubo formula:

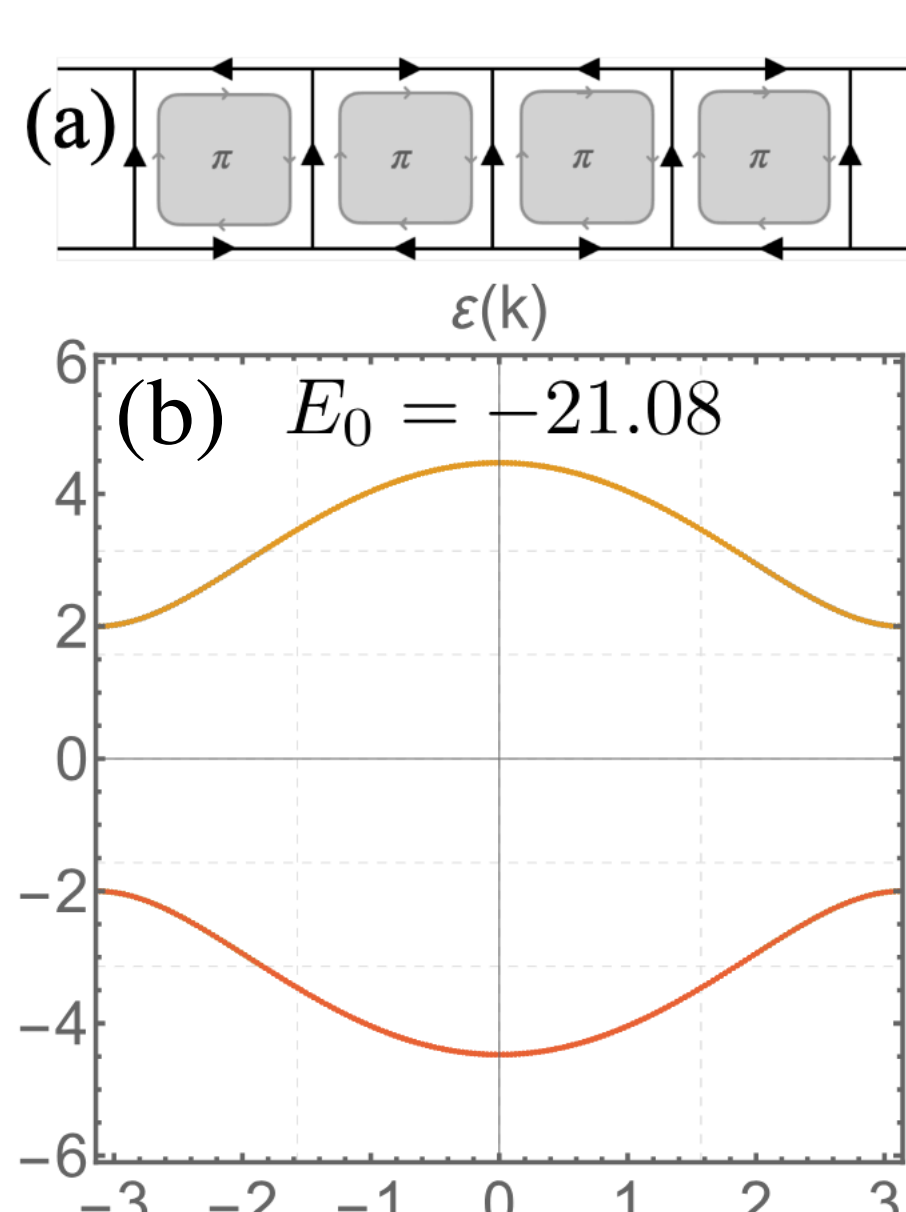
$$\sigma(k, \omega) \sim C_J(k, \omega)/\omega \sim \frac{\omega C_\varepsilon(k, \omega)}{2 - 2 \cos(k)}$$

Result:

- Dynamical energy density response is gapless
- Energy conductivity becomes vanishingly small at low ω



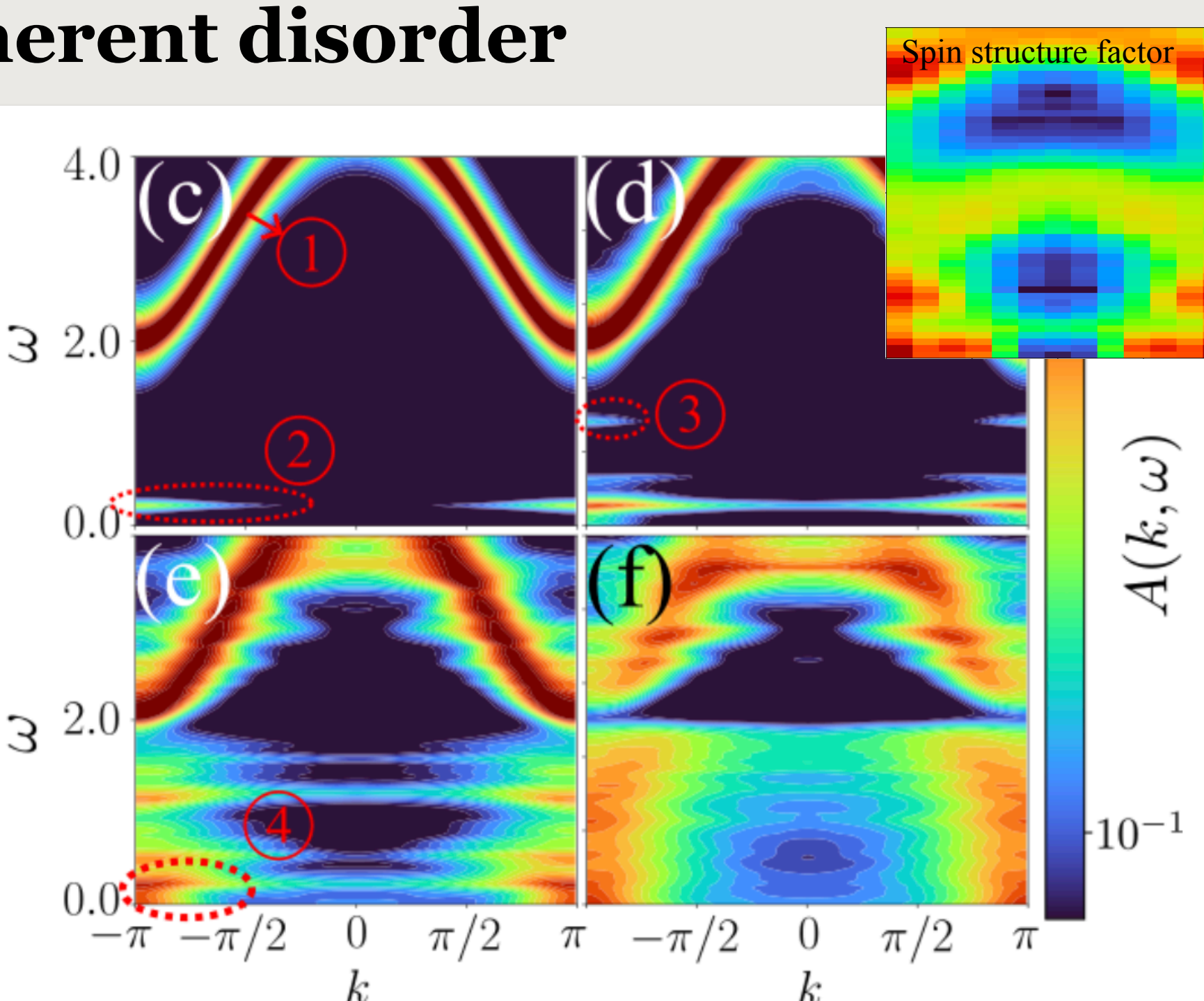
Gapless spectrum by coherent disorder



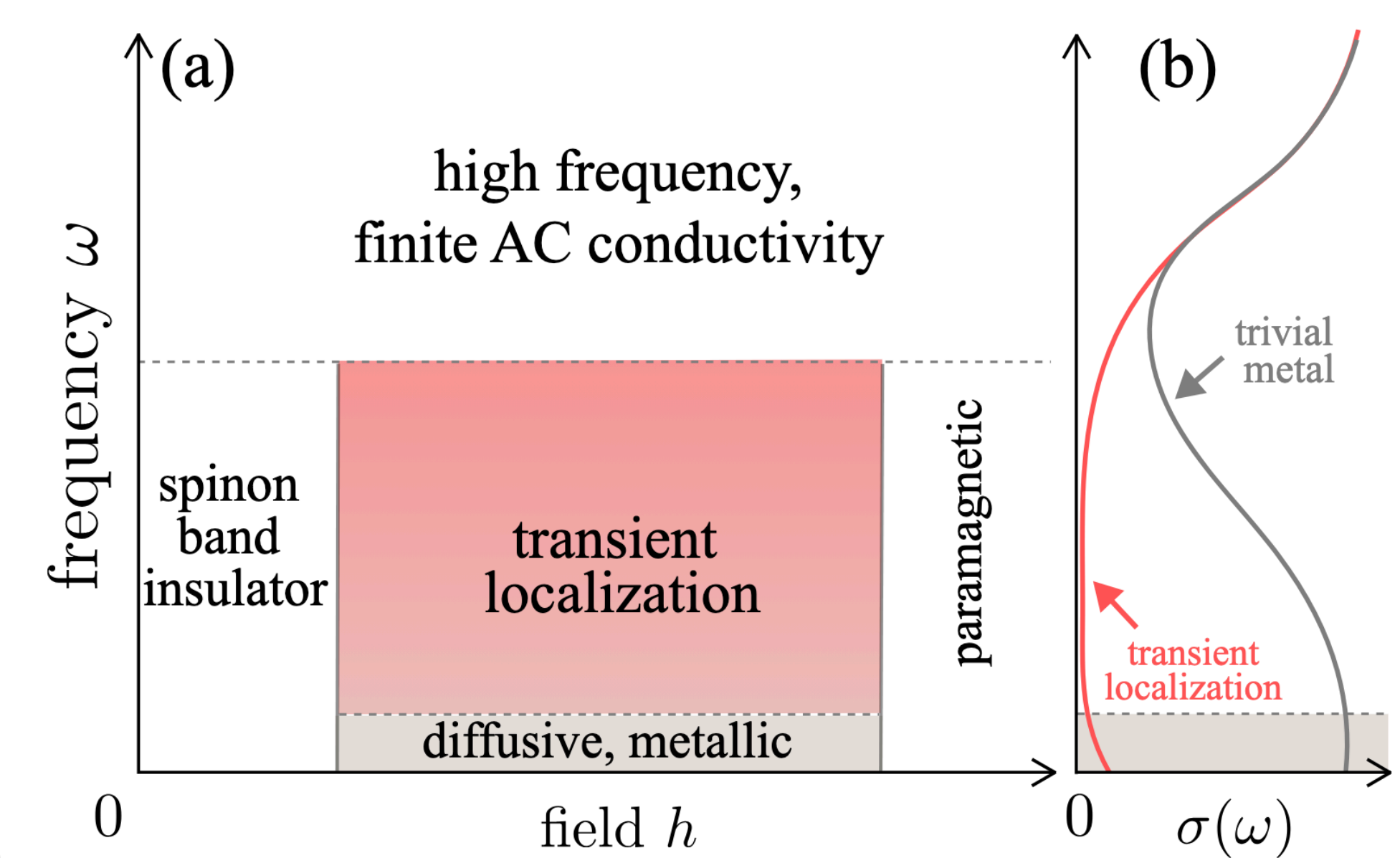
(a,b) At zero field, fermions are gapped band insulator

(c,d) Small field induces low vison density, giving in-gap Majorana resonances.

(e,f) Majorana resonances resolve into gapless continuum at higher field (vison density $\uparrow$)



Intermediate timescales



References

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- P. Zhu*, S. Feng*, K. Wang, T. Xiang, N. Trivedi, Nature Communications 16, 2420 (2025)
- K. Wang*, S. Feng*, P. Zhu, et al., Phys. Rev. B 111, L100402 (2025, Editor's Suggestion)

